



Biometric Fingerprint-Based Electronic Voting System (EVS)

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1st Prize — Genesis of the Trailblazers

UN SDG Goal 9 · Industry, Innovation & Infrastructure

TEAM

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COURSE

10214CS601 — Minor Project-1

1 EXECUTIVE SUMMARY

⚠ PROBLEM

Conventional voting relies on manual voter verification — prone to human error, impersonation, and duplicate voting, undermining electoral integrity and slowing the voting process.

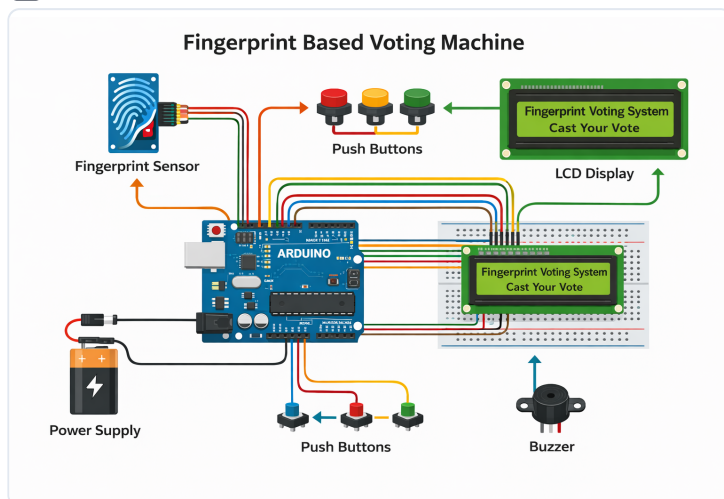
✓ SOLUTION

An autonomous, tamper-proof electronic voting machine that enforces **"One Person – One Vote"** using optical biometric fingerprint authentication to verify each voter before a ballot can be cast.

2 SYSTEM ARCHITECTURE & HARDWARE

COMPONENT	ROLE IN SYSTEM
Arduino Uno (ATmega328P)	Central microcontroller — manages state logic, vote counters, and coordinates all peripherals.
Fingerprint Module (R307 / AS608)	Optical biometric sensor — captures minutiae and communicates with the MCU via UART for template matching.
16×2 LCD (I2C interface)	Real-time status output — "Cast Your Vote", "Voted", "Already Voted", "Admin Mode".
Peripherals	Debounced tactile push buttons for candidate selection; piezo buzzer for audio alerts.

3 ARCHITECTURE DIAGRAM



Circuit-level integration — fingerprint sensor, Arduino Uno, LCD, push buttons, buzzer & power supply

7 ENGINEERING HIGHLIGHTS

01 Low-latency UART link between the optical sensor's onboard DSP and the ATmega328P for real-time minutiae matching.

02 Air-gapped & standalone — operates with zero external internet access, eliminating remote exploit surfaces entirely.

03 Portable, low-power design running on 5V/9V DC, enabling field deployment without fixed infrastructure.

4 WORKFLOW & VERIFICATION LOGIC

A Admin Enrollment

Master verification enrolls and indexes voter fingerprint templates into internal memory, each initialised with `HasVoted = FALSE`.

B Authentication

Voter scans finger; the sensor's DSP matcher checks the live scan against stored minutiae templates.

C Vote Polling

A valid, registered voter casts their vote via push button; the `HasVoted` flag toggles to `TRUE`.

✗ Fraud Lockout

Any duplicate attempt is instantly blocked with an audible buzzer alert and an on-screen "Already Voted" message.

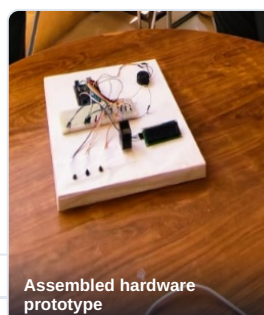
5 EXPERIMENTAL RESULTS

METRIC	MANUAL CHECK	BIOMETRIC EVS
Voter Verification	Moderate accuracy	High accuracy
Duplicate Voting	Hard to detect	100% blocked
Vote Recording	Manual, slow	Instant & automated
Result Generation	Delayed counting	Instant display

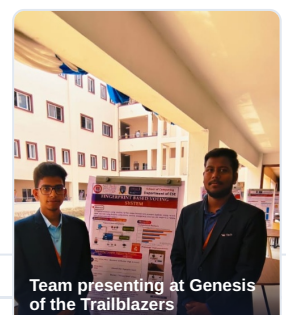
1.20 → 3.40
Efficiency / Security
index across 4 test attempts

100%
Duplicate-vote
attempts blocked

6 PROTOTYPE & FIELD DEMO



Assembled hardware prototype



Team presenting at Genesis of the Trailblazers